Muhammad Sohail Danish

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EXPERIENCE

Student Researcher — **Switzerland + Remote**
Google – Geo — *Apr 2026 – Present*

- Develop large-scale generative models for synthesizing vehicle navigation trajectories conditioned on satellite imagery.
- Formulate spatial and autoregressive models to predict trajectory location, heading, and speed distributions.
- Investigate map freshness using trajectory likelihoods to identify inconsistencies between predicted vehicle behavior and existing road maps.

PhD Resident Fellow — **Kenya**
Microsoft – AI for Good Lab — *Aug 2025 – Feb 2026*

- Led the development of GeoFaham, a multi-agent LLM system for geospatial disaster analysis using satellite imagery, vector data, and specialized tools.
- Developed GeoFaham-Bench, a 40-task benchmark evaluating multi-step geospatial reasoning, planning, and tool use.

PhD Researcher — **United Arab Emirates**
Mohamed bin Zayed University of Artificial Intelligence — *Aug 2023 – Present*

- TerraFM (ICLR 2026): Developed a scalable self-supervised Earth observation foundation model pretrained on 18.7M global Sentinel-1/2 samples, introducing cross-attention fusion and dual-centering for multisensor learning.
- GEOBench-VLM (ICCV 2025 Highlight): Built a geospatial evaluation suite of 10,000+ verified instructions across 31 tasks to benchmark 20+ vision-language models.
- GeoChat (CVPR 2024): Developed a grounded VLM for remote sensing, curating 318K instruction-following samples for image VQA, region captioning, and object grounding.
- EarthDial (CVPR 2025): Contributed to a multisensor, multitemporal conversational VLM trained on 11.1M+ samples.

Graduate Research Assistant — **United Arab Emirates**
Intelligent Visual Analytics Lab — *Sep 2022 – Aug 2023*

- DivAlign (CVPR 2024): Developed a detector-agnostic single-domain generalization method for object detection using data diversification and alignment losses to improve performance on unseen domains.
- Feature-Mode Perturbation (DICTA 2024 Oral): Proposed a feature-space perturbation method to reduce source-domain bias and improve cross-domain detection robustness.

Graduate Student Researcher — **Pakistan**
Intelligent Machine Lab - Information Technology University — *Sep 2020 – Aug 2022*

- Low-Cost Malaria Detection (CVPR 2022): Developed a domain adaptation method for detecting and classifying malaria cells across different microscopy setups.
- Worked on scene graph generation and video face recognition using vision transformers.
- Teaching: Teaching Assistant for Deep Learning (Spring '22), Machine Learning (Fall '21), and Compiler Construction (Fall '20).

PUBLICATIONS

(* indicates authors contributed equally)

- **Muhammad Sohail Danish**, Gilles Hacheme, Akram Zaytar, Satej Soman, Inbal Becker-Reshef, Juan Lavista Ferres, Salman Khan, and Girmaw Abebe Tadesse. "GeoFaham: Domain-Specialized Multi-Agent Geospatial Analytics for Disaster Response". Preprint in preparation, 2026.
- **Muhammad Sohail Danish**, Muhammad Akhtar Munir, Syed Roshaan Ali Shah, Muhammad Haris Khan, Rao Muhammad Anwer, Jorma Laaksonen, Fahad Shahbaz Khan, and Salman Khan. "TerraFM: A Scalable Foundation Model for Unified Multisensor Earth Observation". **ICLR 2026**.
- **Muhammad Sohail Danish***, Muhammad Akhtar Munir*, Syed Roshaan Ali Shah, Kartik Kuckreja, Fahad Shahbaz Khan, Paolo Fraccaro, Alexandre Lacoste, and Salman Khan. "*GEOBench-VLM: Benchmarking Vision-Language Models for Geospatial Tasks.*" **ICCV 2025 (Highlight)**.
- Soni, Sagar, Akshay Dudhane, Hiyam Debary, Mustansar Fiaz, Muhammad Akhtar Munir, **Muhammad Sohail Danish**, Paolo Fraccaro et al. "*Earthdial: Turning multi-sensory earth observations to interactive dialogues.*" In **CVPR 2025**.
- **Muhammad Sohail Danish***, Kuckreja Kartik*, Muzammal Naseer, Abhijit Das, Salman Khan, and Fahad Shahbaz Khan. "*Geochat: Grounded large vision-language model for remote sensing.*" In **CVPR 2024**.
- **Muhammad Sohail Danish**, Muhammad Haris Khan, Muhammad Akhtar Munir, M. Saquib Sarfraz, and Mohsen Ali. "*Improving Single Domain-Generalized Object Detection: A Focus on Diversification and Alignment.*" In **CVPR 2024**

- **Muhammad Sohail Danish**, Javed Iqbal, Mohsen Ali, M. Saquib Sarfraz, Salman Khan, and Muhammad Haris Khan. "*Perturbing Dominant Feature Modes for Single Domain-Generalized Object Detection.*". In **DICTA 2024 (ORAL)**.
- Waqas Sultani, **Muhammad Sohail Danish***, Wajahat Nawaz*, Syed Javed*, Asma Saadia, and Mohsen Ali. "*Towards low-cost and efficient malaria detection.*". In **CVPR 2022.**

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence — **United Arab Emirates**
PhD in Computer Vision | GPA: 3.57/4.00 — *Aug 2023 – Present*

- Recipient of a *fully funded scholarship* sponsored by Govt of the UAE
- *Selected Courses:* Advanced Computer Vision, Advanced 3D Computer Vision, Advanced Techniques in Visual Object Recognition and Detection, Life-long Learning Agents for Vision, Advanced Topics in Vision and Language
- *Research Focus:* Geospatial Foundation Models, Vision-Language Models for Earth Observation
- *Advisor:* **Prof. Salman Khan**, *Co-Advisor:* **Prof. Muhammad Haris Khan**

Information Technology University — **Pakistan**
MS in Data Science | GPA: 3.47/4.00 — *Sept 2020 – Jun 2023*

- Recipient of *Graduate Student Fellowship* for two years
- *Selected Courses:* Mathematical Foundations of Data Science, Machine Learning, Deep Learning, Computer Vision, Big Data Analytics, Information Retrieval and Text Mining
- *MS Thesis:* Single-Domain Generalization for object detection
- *Advisor:* **Prof. Mohsen Ali**, *Co-Advisor:* **Prof. Muhammad Haris Khan**

Qurtuba University of Science and Information Technology — **Pakistan**
BS in Computer Science | GPA: 3.97/4.00 — *Sept 2015 – Jun 2019*

- *Final Year Project:* Luccy, An Autonomous Drone
- *Selected Courses:* Object-Oriented Programming, Data Structures and Algorithms, Artificial Intelligence, Software Engineering, Mobile App Development

TECHNICAL SKILLS

Domains: Computer Vision, Vision-Language Models (VLMs), Generative Modeling, Self-Supervised Foundation Models, Earth Observation & Geospatial AI, Multi-Agent Systems, Domain Generalization & Adaptation
Libraries: PyTorch, JAX, Flax, TensorFlow, Hugging Face Transformers, MMDetection, AutoGen, OpenCV, Scikit-learn
Languages & Geospatial/Systems Stack: Python, SQL, PHP, JavaScript, GDAL/Rasterio, PostGIS, TiTiler, FastAPI, Docker, Linux, Multi-GPU & TPU Distributed Training

References can be provided upon request.